1. Description

1.1. Project

Project Name	STM32H750B-DK
Board Name	custom
Generated with:	STM32CubeMX 6.15.0
Date	12/30/2025

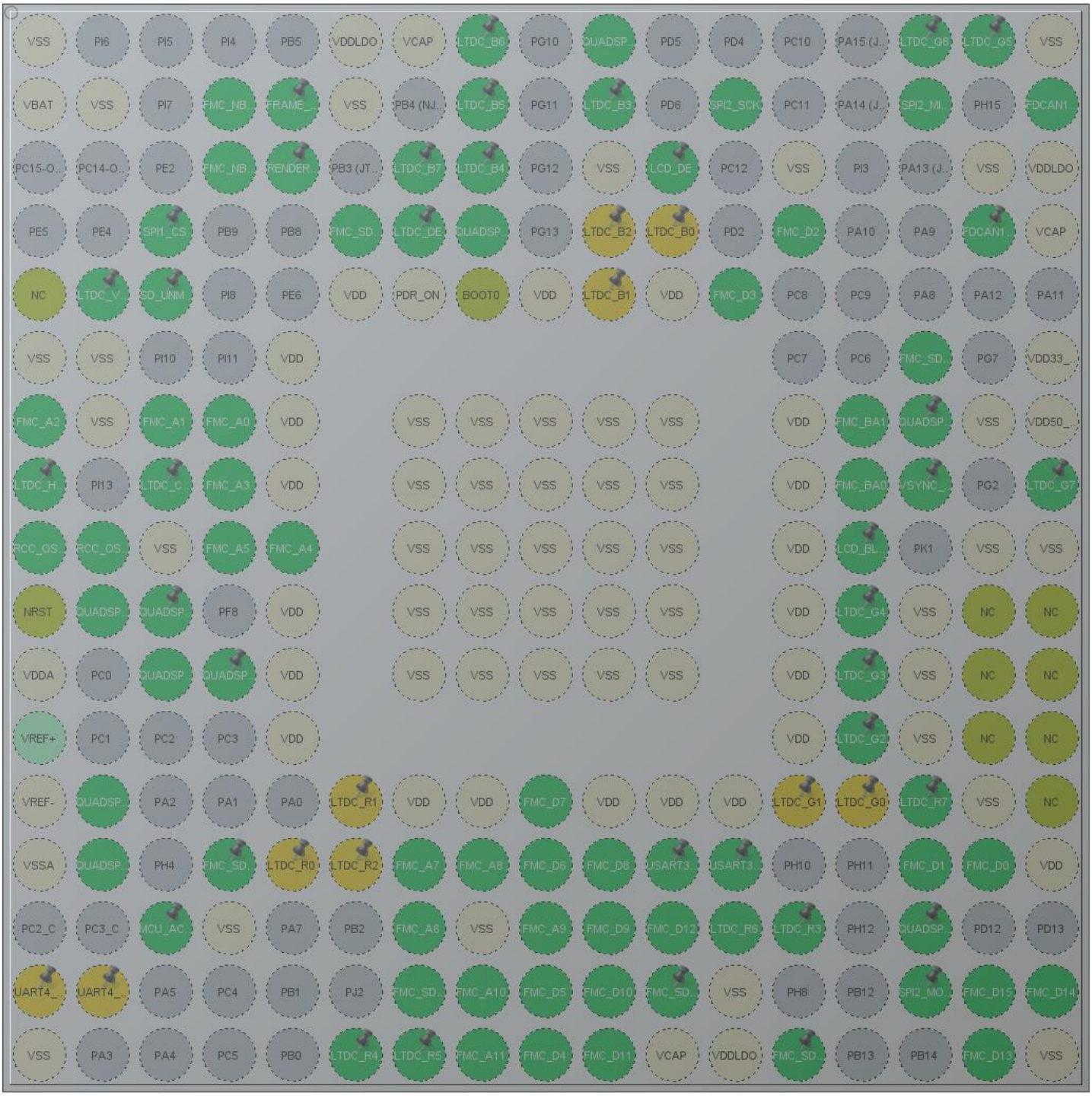
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H750 Value line
MCU name	STM32H750XBHx
MCU Package	TFBGA240
MCU Pin number	265

1.3. Core(s) information

Core(s)	ARM Cortex-M7

2. Pinout Configuration



TFBGA240 +25 (Top view)

3. Pins Configuration

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	VSS	Power		
A6	VDDLDO	Power		
A7	VCAP	Power		
A8	PK5	I/O	LTDC_B6	
A10	PG9	I/O	QUADSPI_BK2_IO2	
A15	PI1	I/O	LTDC_G6	
A16	PI0	I/O	LTDC_G5	
A17	VSS	Power		
B1	VBAT	Power		
B2	VSS	Power		
B4	PE1	I/O	FMC_NBL1	
B5	PB6 *	I/O	GPIO_Output	FRAME_RATE
B6	VSS	Power		
B8	PK4	I/O	LTDC_B5	
B10	PJ15	I/O	LTDC_B3	
B12	PD3	I/O	SPI2_SCK	
B15	PI2	I/O	SPI2_MISO	
B17	PH14	I/O	FDCAN1_RX	
C4	PE0	I/O	FMC_NBL0	
C5	PB7 *	I/O	GPIO_Output	RENDER_TIME
C7	PK6	I/O	LTDC_B7	
C8	PK3	I/O	LTDC_B4	
C10	VSS	Power		
C11	PD7 *	I/O	GPIO_Output	LCD_DE
C13	VSS	Power		
C16	VSS	Power		
C17	VDDLDO	Power		
D3	PE3 *	I/O	GPIO_Output	SPI1_CS
D6	PG15	I/O	FMC_SDNCAS	
D7	PK7	I/O	LTDC_DE	
D8	PG14	I/O	QUADSPI_BK2_IO3	
D10	PJ14 **	I/O	LTDC_B2	
D11	PJ12 **	I/O	LTDC_B0	
D13	PD0	I/O	FMC_D2	
D16	PH13	I/O	FDCAN1_TX	
D17	VCAP	Power		

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
E1	NC	NC		
E2	PI9	I/O	LTDC_VSYNC	
E3	PC13	I/O	GPIO_EXTI13	SD_UNMOUNT_BTN
E6	VDD	Power		
E7	PDR_ON	Power		
E8	BOOT0	Boot		
E9	VDD	Power		
E10	PJ13 **	I/O	LTDC_B1	
E11	VDD	Power		
E12	PD1	I/O	FMC_D3	
F1	VSS	Power		
F2	VSS	Power		
F5	VDD	Power		
F15	PG8	I/O	FMC_SDCLK	
F17	VDD33_USB	Power		
G1	PF2	I/O	FMC_A2	
G2	VSS	Power		
G3	PF1	I/O	FMC_A1	
G4	PF0	I/O	FMC_A0	
G5	VDD	Power		
G7	VSS	Power		
G8	VSS	Power		
G9	VSS	Power		
G10	VSS	Power		
G11	VSS	Power		
G13	VDD	Power		
G14	PG5	I/O	FMC_BA1	
G15	PG6	I/O	QUADSPI_BK1_NCS	
G16	VSS	Power		
G17	VDD50_USB	Power		
H1	PI12	I/O	LTDC_HSYNC	
H3	PI14	I/O	LTDC_CLK	
H4	PF3	I/O	FMC_A3	
H5	VDD	Power		
H7	VSS	Power		
H8	VSS	Power		
H9	VSS	Power		
H10	VSS	Power		
H11	VSS	Power		

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
H13	VDD	Power		
H14	PG4	I/O	FMC_BA0	
H15	PG3 *	I/O	GPIO_Output	VSYNC_FREQ
H17	PK2	I/O	LTDC_G7	
J1	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
J2	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
J3	VSS	Power		
J4	PF5	I/O	FMC_A5	
J5	PF4	I/O	FMC_A4	
J7	VSS	Power		
J8	VSS	Power		
J9	VSS	Power		
J10	VSS	Power		
J11	VSS	Power		
J13	VDD	Power		
J14	PK0 *	I/O	GPIO_Output	LCD_BL_CTRL
J16	VSS	Power		
J17	VSS	Power		
K1	NRST	Reset		
K2	PF6	I/O	QUADSPI_BK1_IO3	
K3	PF7	I/O	QUADSPI_BK1_IO2	
K5	VDD	Power		
K7	VSS	Power		
K8	VSS	Power		
K9	VSS	Power		
K10	VSS	Power		
K11	VSS	Power		
K13	VDD	Power		
K14	PJ11	I/O	LTDC_G4	
K15	VSS	Power		
K16	NC	NC		
K17	NC	NC		
L1	VDDA	Power		
L3	PF10	I/O	QUADSPI_CLK	
L4	PF9	I/O	QUADSPI_BK1_IO1	
L5	VDD	Power		
L7	VSS	Power		
L8	VSS	Power		
L9	VSS	Power		

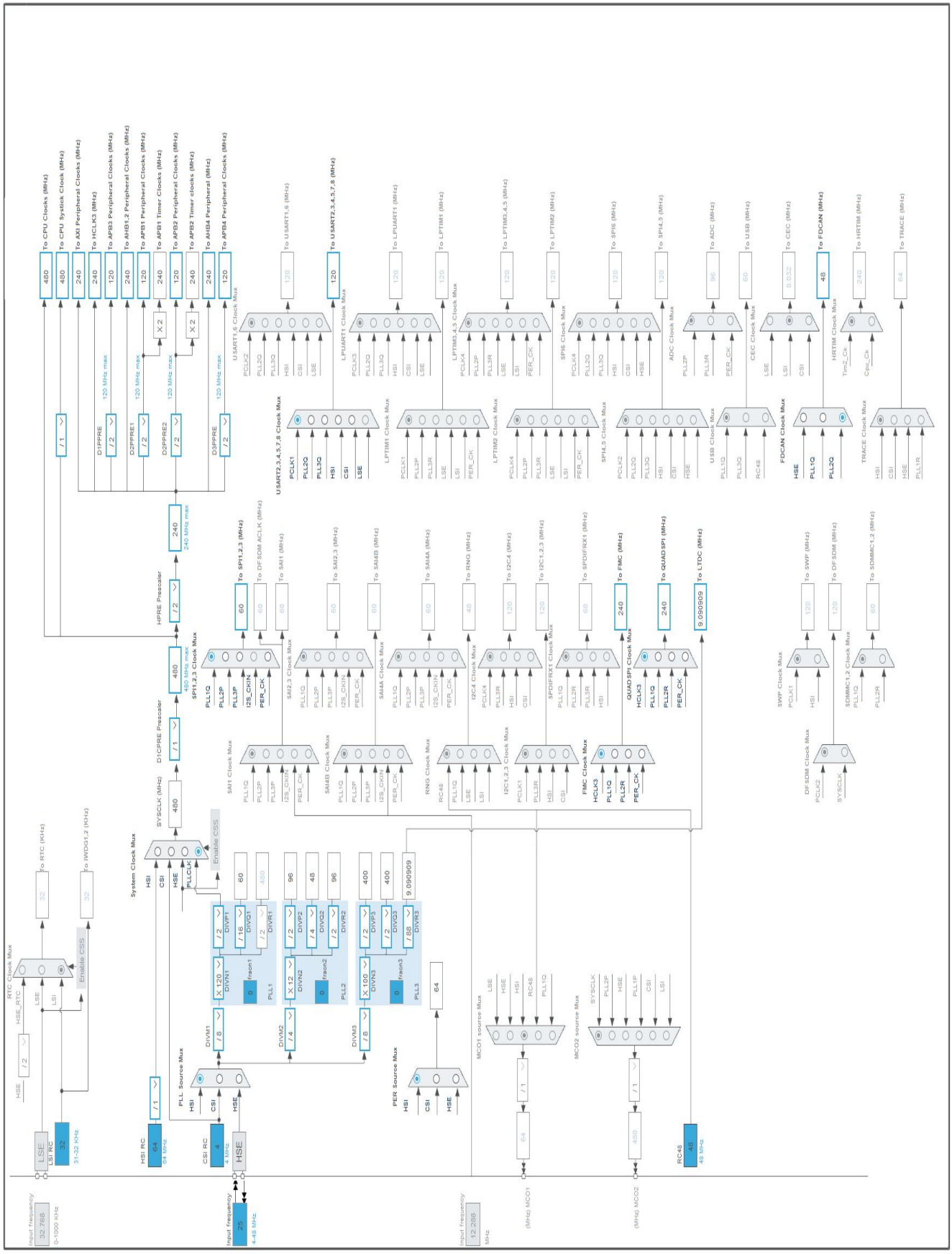
Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
L10	VSS	Power		
L11	VSS	Power		
L13	VDD	Power		
L14	PJ10	I/O	LTDC_G3	
L15	VSS	Power		
L16	NC	NC		
L17	NC	NC		
M5	VDD	Power		
M13	VDD	Power		
M14	PJ9	I/O	LTDC_G2	
M15	VSS	Power		
M16	NC	NC		
M17	NC	NC		
N1	VREF-	Power		
N2	PH2	I/O	QUADSPI_BK2_IO0	
N6	PJ0 **	I/O	LTDC_R1	
N7	VDD	Power		
N8	VDD	Power		
N9	PE10	I/O	FMC_D7	
N10	VDD	Power		
N11	VDD	Power		
N12	VDD	Power		
N13	PJ8 **	I/O	LTDC_G1	
N14	PJ7 **	I/O	LTDC_G0	
N15	PJ6	I/O	LTDC_R7	
N16	VSS	Power		
N17	NC	NC		
P1	VSSA	Power		
P2	PH3	I/O	QUADSPI_BK2_IO1	
P4	PH5	I/O	FMC_SDNWE	
P5	PI15 **	I/O	LTDC_R0	
P6	PJ1 **	I/O	LTDC_R2	
P7	PF13	I/O	FMC_A7	
P8	PF14	I/O	FMC_A8	
P9	PE9	I/O	FMC_D6	
P10	PE11	I/O	FMC_D8	
P11	PB10	I/O	USART3_TX	
P12	PB11	I/O	USART3_RX	
P15	PD15	I/O	FMC_D1	

Pin Number TFBGA240	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
P16	PD14	I/O	FMC_D0	
P17	VDD	Power		
R3	PA6 *	I/O	GPIO_Output	MCU_ACTIVE
R4	VSS	Power		
R7	PF12	I/O	FMC_A6	
R8	VSS	Power		
R9	PF15	I/O	FMC_A9	
R10	PE12	I/O	FMC_D9	
R11	PE15	I/O	FMC_D12	
R12	PJ5	I/O	LTDC_R6	
R13	PH9	I/O	LTDC_R3	
R15	PD11	I/O	QUADSPI_BK1_IO0	
T1	PA0_C **	I/O	UART4_TX	
T2	PA1_C **	I/O	UART4_RX	
T7	PF11	I/O	FMC_SDNRAS	
T8	PG0	I/O	FMC_A10	
T9	PE8	I/O	FMC_D5	
T10	PE13	I/O	FMC_D10	
T11	PH6	I/O	FMC_SDNE1	
T12	VSS	Power		
T15	PB15	I/O	SPI2_MOSI	
T16	PD10	I/O	FMC_D15	
T17	PD9	I/O	FMC_D14	
U1	VSS	Power		
U6	PJ3	I/O	LTDC_R4	
U7	PJ4	I/O	LTDC_R5	
U8	PG1	I/O	FMC_A11	
U9	PE7	I/O	FMC_D4	
U10	PE14	I/O	FMC_D11	
U11	VCAP	Power		
U12	VDDLDO	Power		
U13	PH7	I/O	FMC_SDCKE1	
U16	PD8	I/O	FMC_D13	
U17	VSS	Power		

* The pin is affected with an I/O function

** The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H750 Value line
MCU	STM32H750XBHx
Datasheet	DS12556_Rev6

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

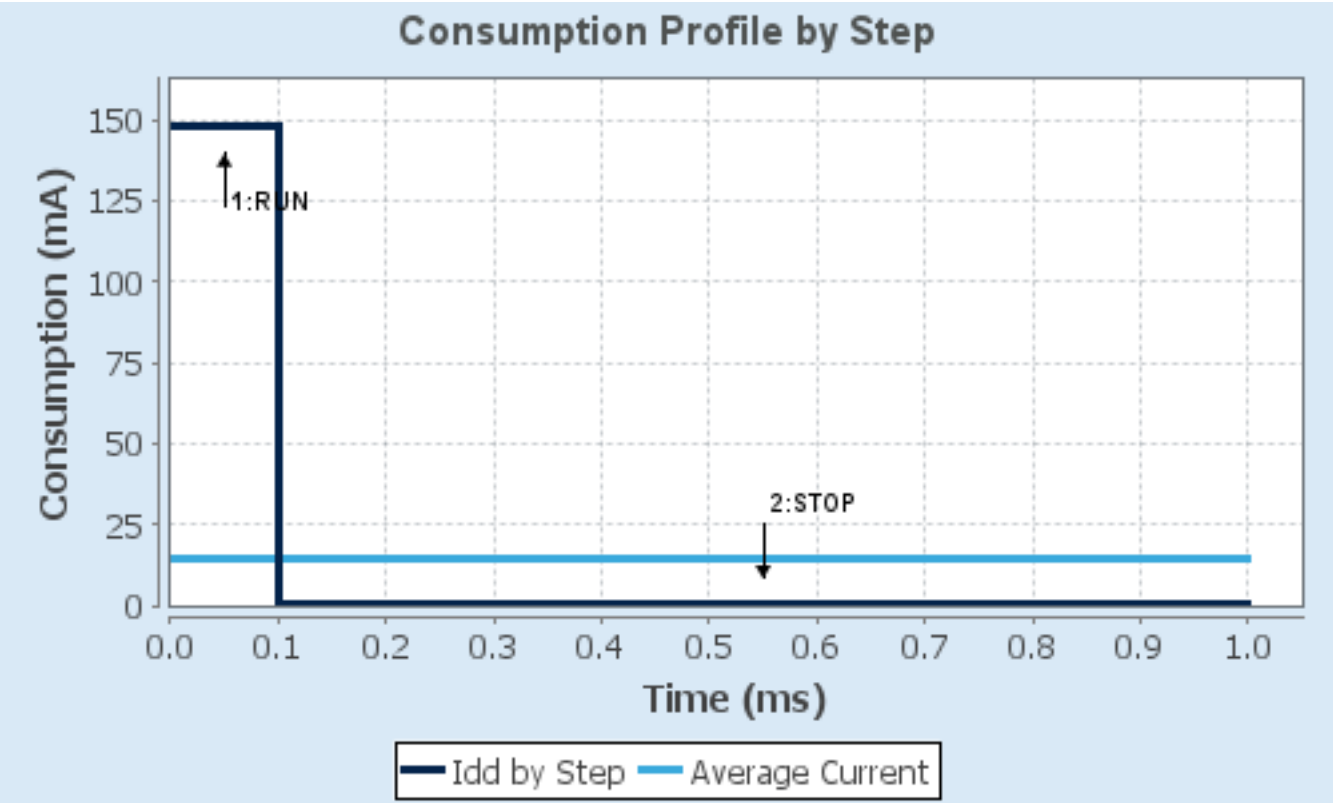
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0-High	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	ITCM	NA
CPU Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL	Flash-OFF
Clock Source Frequency	24 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	148 mA	150 µA
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Ta Max	108.57	124.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	14.94 mA
Battery Life	1 day, 17 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
150
125
100
75
50
25
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	STM32H750B-DK
Project Folder	C:\TouchGFXProjects\act_dashboard
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_H7 V1.12.1
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x1000
Minimum Stack Size	0x1000

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	SystemClock_Config	RCC
3	MX_MDMA_Init	MDMA
4	MX_LTDC_Init	LTDC
5	MX_DMA2D_Init	DMA2D
6	MX_QUADSPI_Init	QUADSPI
7	MX_FMC_Init	FMC
8	MX_LIBJPEG_Init	LIBJPEG
9	MX_JPEG_Init	JPEG
10	MX_CRC_Init	CRC
11	MX_FDCAN1_Init	FDCAN1

Rank	Function Name	Peripheral Instance Name
12	MX_SPI2_Init	SPI2
13	MX_FATFS_Init	FATFS
14	MX_USART3_UART_Init	USART3
16	MX_TouchGFX_Init	STMicroelectronics.X-CUBE-TOUCHGFX.4.25.0
17	MX_TouchGFX_Process	STMicroelectronics.X-CUBE-TOUCHGFX.4.25.0

3. Peripherals and Middlewares Configuration

3.1. CORTEX_M7

3.1.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode	Disabled

Cortex Interface Settings:

CPU ICache	**Enabled ***
CPU DCache	**Enabled ***

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	**Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers ***

Cortex Memory Protection Unit Region 0 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x24000000 ***
MPU Region Size	**512KB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	**DISABLE ***
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	**ENABLE ***
MPU Bufferable Permission	**ENABLE ***

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x90000000 ***
MPU Region Size	**256MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS NOT PERMITTED
MPU Instruction Access	**DISABLE ***
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 2 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x90000000 ***

MPU Region Size	**128MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	ENABLE
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	**ENABLE ***
MPU Bufferable Permission	**ENABLE ***

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0xD0000000 ***
MPU Region Size	**256MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS NOT PERMITTED
MPU Instruction Access	**DISABLE ***
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0xD0000000 ***
MPU Region Size	**32MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	level 0
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	**DISABLE ***
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	**ENABLE ***
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region	Disabled

3.2. CRC

mode: Activated

3.2.1. Parameter Settings:

Basic Parameters:

Default Polynomial State	Enable
Default Init Value State	Enable

Advanced Parameters:

Input Data Inversion Mode	None
Output Data Inversion Mode	Disable
Input Data Format	Bytes

3.3. DMA2D

mode: Activated

3.3.1. Parameter Settings:

Basic Parameters:

Transfer Mode	Memory to Memory
Color Mode	**RGB565 ***
Output Offset	0

Foreground layer Configuration:

DMA2D Input Color Mode	RGB565
DMA2D ALPHA MODE	No modification of the alpha channel value
Input Alpha	0
Input Offset	0
DMA2D ALPHA Inversion	Regular Alpha
DMA2D Red and Blue swap	Regular mode (RGB or ARGB)
DMA2D Chroma Sub-Sampling Mode	No chroma sub-sampling 4:4:4

3.4. FDCAN1

mode: Activated

3.4.1. Parameter Settings:

Basic Parameters:

Frame Format	**FD mode without BitRate Switching ***
Mode	Normal mode
Auto Retransmission	**Enable ***
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	**12 ***
Data Prescaler	**24 ***
Data Sync Jump Width	1
Data Time Seg1	**2 ***
Data Time Seg2	1
Message Ram Offset	0
Std Filters Nbr	**1 ***
Ext Filters Nbr	0
Rx Fifo0 Elmts Nbr	**1 ***
Rx Fifo0 Elmt Size	8 bytes data field
Rx Fifo1 Elmts Nbr	0
Rx Fifo1 Elmt Size	8 bytes data field
Rx Buffers Nbr	0
Rx Buffer Size	8 bytes data field
Tx Events Nbr	0
Tx Buffers Nbr	0
Tx Fifo Queue Elmts Nbr	**1 ***
Tx Fifo Queue Mode	FIFO mode
Tx Elmt Size	8 bytes data field

Clock Calibration Unit:

Clock Calibration	Disable

Bit Timings Parameters:

Nominal Prescaler	**8 ***
Nominal Time Quantum	**166.66666666666666 ***
Nominal Time Seg1	**83 ***
Nominal Time Seg2	**12 ***
Nominal Time for one Bit	**16000 ***
Nominal Baud Rate	**62500 ***

3.5. FMC

SDRAM 2

Clock and chip enable: SDCKE1+SDNE1

Internal bank number: 4 banks

Address: 12 bits

Data: 16 bits

Byte enable: 16-bit byte enable

3.5.1. SDRAM 2:

SDRAM control:

Bank	SDRAM bank 2
Number of column address bits	8 bits
Number of row address bits	12 bits
CAS latency	**3 memory clock cycles ***
Write protection	Disabled
SDRAM common clock	**2 HCLK clock cycles ***
SDRAM common burst read	**Enabled ***
SDRAM common read pipe delay	0 HCLK clock cycle

SDRAM timing in memory clock cycles:

Load mode register to active delay	**2 ***
Exit self-refresh delay	**7 ***
Self-refresh time	**4 ***
SDRAM common row cycle delay	**7 ***
Write recovery time	**5 ***
SDRAM common row precharge delay	**2 ***
Row to column delay	**2 ***

3.5.2. Bank Mapping:

Mapping parameters:

FMC bank mapping	Default mapping

3.6. JPEG
mode: Activated
3.6.1. Parameter Settings:

Version:

JPEG version	jpeg1_v1_0

JPEG Software options:

ENCODE	Enabled
DECODE	Enabled
RGB_FORMAT	**JPEG_RGB565 ***
JPEG_SWAP_RG	0

3.7. LTDC
Display Type: RGB565 (16 bits)
3.7.1. Parameter Settings:

Synchronization for Width:

Horizontal Synchronization Width	**40 ***
Horizontal Back Porch	**3 ***
Active Width	**480 ***
Horizontal Front Porch	6
HSync Width	39
Accumulated Horizontal Back Porch Width	42
Accumulated Active Width	522
Total Width	528

Synchronization for Height:

Vertical Synchronization Height	**9 ***
Vertical Back Porch	**3 ***
Active Height	**272 ***
Vertical Front Porch	2
VSync Height	8

Accumulated Vertical Back Porch Height	11
Accumulated Active Height	283
Total Height	285

Signal Polarity:

Horizontal Synchronization Polarity	Active Low
Vertical Synchronization Polarity	Active Low
Data Enable Polarity	Active Low
Pixel Clock Polarity	Normal Input

Layer Default Color:

Red	0
Green	0
Blue	0

3.7.2. Layer Settings:

Layer Default Color:

Layer 0 - Alpha	0
Layer 0 - Blue	0
Layer 0 - Green	0
Layer 0 - Red	0

Windows Position:

Layer 0 - Window Horizontal Start	0
Layer 0 - Window Horizontal Stop	**480 ***
Layer 0 - Window Vertical Start	0
Layer 0 - Window Vertical Stop	**272 ***

Pixel Parameters:

Layer 0 - Pixel Format	**RGB565 ***

Blending:

Layer 0 - Alpha constant for blending	**255 ***
Layer 0 - Blending Factor1	Alpha constant
Layer 0 - Blending Factor2	Alpha constant

Frame Buffer:

Layer 0 - Color Frame Buffer Start Adress	0
Layer 0 - Color Frame Buffer Line Length (Image Width)	**480 ***
Layer 0 - Color Frame Buffer Number of Lines (Image Height)	**272 ***

Number of Layers:

Number of Layers	**1 layer ***

3.8. MEMORYMAP

mode: Activated

3.9. QUADSPI

QuadSPI Mode: Dual Bank with Quad Lines

Chip Select for Dual bank: Enable Chip Select 1 for both banks

3.9.1. Parameter Settings:

General Parameters:

Clock Prescaler	**3 ***
Fifo Threshold	1
Sample Shifting	No Sample Shifting
Flash Size	**26 ***
Chip Select High Time	**4 Cycles ***
Clock Mode	Low
Dual Flash	Enabled

3.10. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.10.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	4 WS (5 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 8 and 16 MHz
PLL2 input frequency range	Between 8 and 16 MHz
PLL3 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	Wide VCO range
PLL2 clock Output range	Wide VCO range
PLL3 clock Output range	Wide VCO range

3.11. SPI2

Mode: Full-Duplex Master

3.11.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	**8 Bits ***
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	**256 ***
Baud Rate	**234.375 KBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.12. SYS

Timebase Source: TIM6

3.13. USART3

Mode: Asynchronous

3.13.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.14. FATFS

mode: User-defined

3.14.1. Set Defines:

Version:

FATFS version	R0.12c

Function Parameters:

FS_READONLY (Read-only mode)	Disabled
FS_MINIMIZE (Minimization level)	Disabled
USE_STRFUNC (String functions)	Enabled with LF -> CRLF conversion
USE_FIND (Find functions)	Disabled

USE_MKFS (Make filesystem function)	Enabled
USE_FASTSEEK (Fast seek function)	Enabled
USE_EXPAND (Use f_expand function)	Disabled
USE_CHMOD (Change attributes function)	Disabled
USE_LABEL (Volume label functions)	Disabled
USE_FORWARD (Forward function)	Disabled

Locale and Namespace Parameters:

CODE_PAGE (Code page on target)	Latin 1
USE_LFN (Use Long Filename)	Disabled
MAX_LFN (Max Long Filename)	255
LFN_UNICODE (Enable Unicode)	ANSI/OEM
STRF_ENCODE (Character encoding)	UTF-8
FS_RPATH (Relative Path)	Disabled

Physical Drive Parameters:

VOLUMES (Logical drives)	1
MAX_SS (Maximum Sector Size)	512
MIN_SS (Minimum Sector Size)	512
MULTI_PARTITION (Volume partitions feature)	Disabled
USE_TRIM (Erase feature)	Disabled
FS_NOFSINFO (Force full FAT scan)	0

System Parameters:

FS_TINY (Tiny mode)	Disabled
FS_EXFAT (Support of exFAT file system)	Disabled
FS_NORTC (Timestamp feature)	Dynamic timestamp
FS_REENTRANT (Re-Entrancy)	Enabled
FS_TIMEOUT (Timeout ticks)	1000
USE_MUTEX	Disabled
SYNC_t (O/S sync object)	osSemaphoreId_t
FS_LOCK (Number of files opened simultaneously)	2

3.15. FREERTOS

Interface: CMSIS_V2

3.15.1. Config parameters:

API:

FreeRTOS API	CMSIS v2

Versions:

FreeRTOS version	10.3.1
CMSIS-RTOS version	2.00

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Disabled

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	**Enabled ***
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	**100000 ***
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	**Enabled ***
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2

TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.15.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.15.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	**Enabled ***

Project settings (see parameter description first):

Use FW pack heap file	Enabled

3.16. LIBJPEG
mode: Enabled
3.16.1. Config parameters:

Version:

LIBJPEG version	8d

MW configuration:

Data Stream management type	FatFS
FREERTOS	Enabled
HAVE_BOOLEAN	Undefined

General Settings:

Use FREERTOS Memory Allocator	Enabled

RGB scanline format:

RGB_ORDERING	**BGR ***

3.17. STMicroelectronics.X-CUBE-TOUCHGFX.4.25.0
mode: GraphicsJjApplication
3.17.1. TouchGFX Generator:

Display:

Interface	**Parallel RGB (LTDC) ***
Framebuffer Pixel Format (LTDC)	RGB565
Width (LTDC)	480
Height (LTDC)	272
Use Larger Framebuffer Stride	No
Framebuffer Strategy	**Double Buffer ***
Buffer Location	By Allocation

Driver:

Application Tick Source	**LTDC ***

Use DMA2D Accelerator (ChromART)	**Yes ***
Real-Time Operating System	CMSIS_RTOS_V2
Additional Features:	
External Data Reader	Disabled
Vector Rendering	**Software ***
Vector Font Rendering	**Enabled ***
Video Decoding:	
Type	**Hardware ***
Concurrent videos	1
Strategy	Direct to Framebuffer
Orientation	Native

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
FDCAN1	PH14	FDCAN1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PH13	FDCAN1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
FMC	PE1	FMC_NBL1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE0	FMC_NBL0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG15	FMC_SDNCAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD0	FMC_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD1	FMC_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG8	FMC_SDCLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF2	FMC_A2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF1	FMC_A1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF0	FMC_A0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG5	FMC_BA1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF3	FMC_A3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG4	FMC_BA0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF5	FMC_A5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF4	FMC_A4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE10	FMC_D7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH5	FMC_SDNWE	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF13	FMC_A7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF14	FMC_A8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE9	FMC_D6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE11	FMC_D8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD15	FMC_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD14	FMC_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF12	FMC_A6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF15	FMC_A9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE12	FMC_D9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE15	FMC_D12	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF11	FMC_SDNRAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG0	FMC_A10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE8	FMC_D5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE13	FMC_D10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH6	FMC_SDNE1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD10	FMC_D15	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD9	FMC_D14	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG1	FMC_A11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PE7	FMC_D4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE14	FMC_D11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH7	FMC_SDCKE1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD8	FMC_D13	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
LTDC	PK5	LTDC_B6	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PI1	LTDC_G6	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PI0	LTDC_G5	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PK4	LTDC_B5	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ15	LTDC_B3	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PK6	LTDC_B7	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PK3	LTDC_B4	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PK7	LTDC_DE	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PI9	LTDC_VSYNC	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PI12	LTDC_HSYNC	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PI14	LTDC_CLK	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PK2	LTDC_G7	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ11	LTDC_G4	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ10	LTDC_G3	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ9	LTDC_G2	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ6	LTDC_R7	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ5	LTDC_R6	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PH9	LTDC_R3	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ3	LTDC_R4	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ4	LTDC_R5	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
QUADSPI	PG9	QUADSPI_BK2_IO2	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PG14	QUADSPI_BK2_IO3	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PG6	QUADSPI_BK1_NCS	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PF6	QUADSPI_BK1_IO3	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PF7	QUADSPI_BK1_IO2	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PF10	QUADSPI_CLK	Alternate Function Push Pull	No pull-up and no pull-down	**Very High**	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
					*	
	PF9	QUADSPI_BK1_IO1	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PH2	QUADSPI_BK2_IO0	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PH3	QUADSPI_BK2_IO1	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PD11	QUADSPI_BK1_IO0	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
RCC	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a	
SPI2	PD3	SPI2_SCK	Alternate Function Push Pull	**Pull-down ***	Low	
	PI2	SPI2_MISO	Alternate Function Push Pull	**Pull-up ***	Low	
	PB15	SPI2_MOSI	Alternate Function Push Pull	**Pull-down ***	Low	
USART3	PB10	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB11	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
Single Mapped Signals	PJ14	LTDC_B2	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ12	LTDC_B0	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ13	LTDC_B1	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ0	LTDC_R1	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ8	LTDC_G1	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ7	LTDC_G0	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PI15	LTDC_R0	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PJ1	LTDC_R2	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PA0_C	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA1_C	UART4_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
GPIO	PB6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	FRAME_RATE
	PB7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	RENDER_TIME
	PD7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LCD_DE
	PE3	GPIO_Output	Output Push Pull	**Pull-down ***	Low	SPI1_CS
	PC13	GPIO_EXTI13	**External Interrupt**	No pull-up and no pull-down	n/a	SD_UNMOUNT_BTN

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
			Mode with Falling edge trigger detection			
	PG3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	VSYNC_FREQ
	PK0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LCD_BL_CTRL
	PA6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	**Very High ***	MCU_ACTIVE

4.2. DMA configuration

nothing configured in DMA service

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

MDMA Request	Channel	Trigger Mode	Priotity
JPEG_INFIFO_TH	MDMA_Channel7	Buffer transfer	**High ***
JPEG_OUTFIFO_TH	MDMA_Channel6	Buffer transfer	**Very High ***

JPEG_INFIFO_TH: MDMA_Channel7 DMA request Settings:

Buffer Transfer Length:	**32 ***
Data Mask:	0
Source Burst Size:	**Burst of 32 beats ***
Destination Burst Size:	**Burst of 16 beats ***
Block Data Length:	0
Destination Address:	0
Destination Block Address Offset:	0
Address Mask:	0
Data Alignement:	N/A
Source Block Address Offset:	0
Endianness:	Little endianness preserved
Source Address:	0

Source Increment mode:	Increment by a byte
Block Count:	0
Destination Increment mode:	Fixed
Source Data Width:	Byte
Destination Data Width:	**Word ***

JPEG_OUTFIFO_TH: MDMA_Channel6 DMA request Settings:

Buffer Transfer Length:	**32 ***
Data Mask:	0
Source Burst Size:	**Burst of 32 beats ***
Destination Burst Size:	**Burst of 32 beats ***
Block Data Length:	0
Destination Address:	0
Destination Block Address Offset:	0
Address Mask:	0
Data Alignement:	N/A
Source Block Address Offset:	0
Endianness:	Little endianness preserved
Source Address:	0
Source Increment mode:	Fixed
Block Count:	0
Destination Increment mode:	Increment by a byte
Source Data Width:	**Word ***
Destination Data Width:	Byte

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
FDCAN1 interrupt 0	true	5	0
EXTI line[15:10] interrupts	true	5	0
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	true	0	0
LTDC global interrupt	true	5	0
DMA2D global interrupt	true	5	0
JPEG global interrupt	true	5	0
MDMA global interrupt	true	5	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
FDCAN1 interrupt 1	unused		
SPI2 global interrupt	unused		
USART3 global interrupt	unused		
FMC global interrupt	unused		
FDCAN calibration unit interrupt	unused		
FPU global interrupt	unused		
LTDC global error interrupt	unused		
QUADSPI global interrupt	unused		
HSEM1 global interrupt	unused		

4.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false

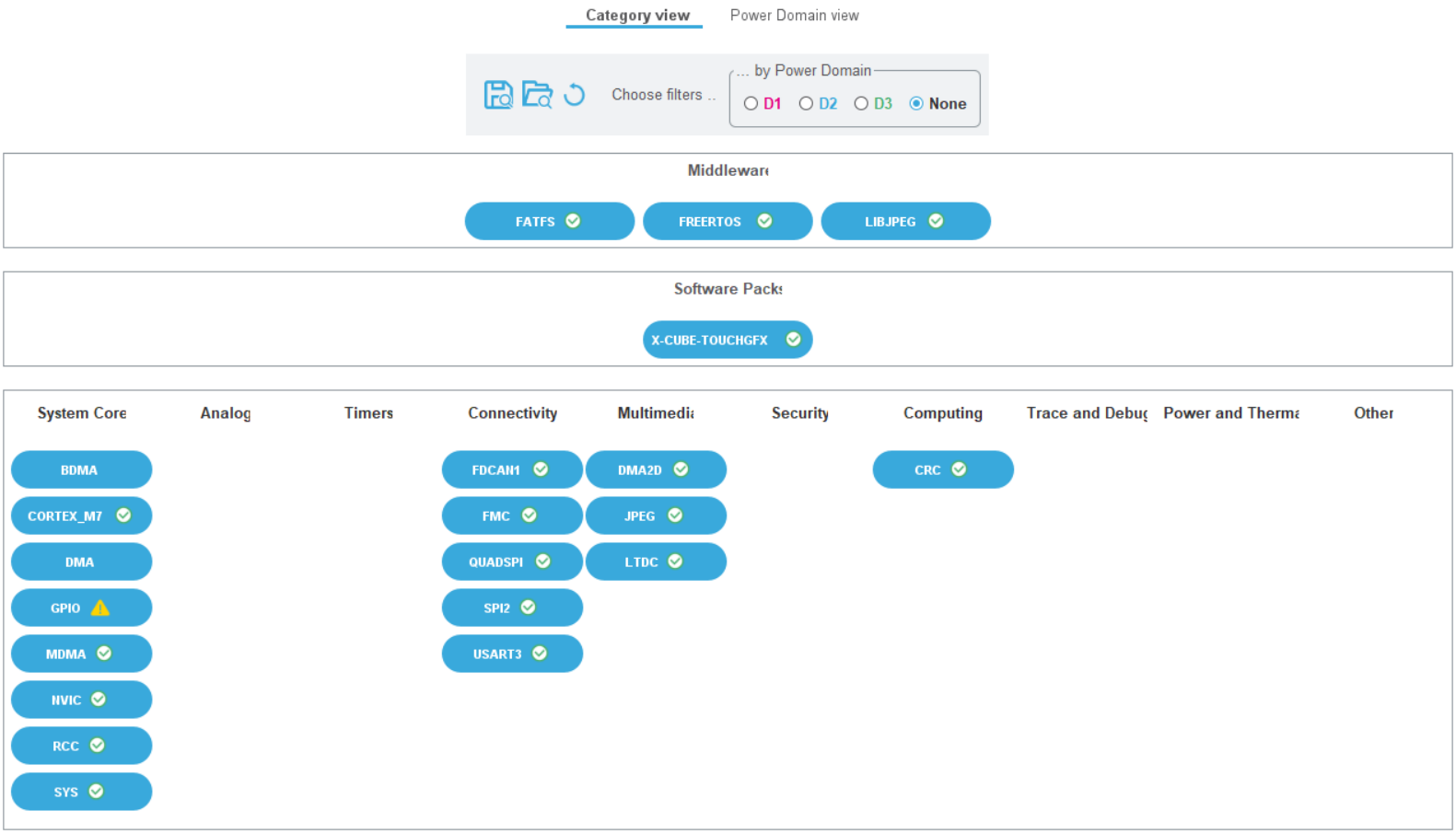
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
FDCAN1 interrupt 0	false	true	true
EXTI line[15:10] interrupts	false	true	true
TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts	false	true	true
LTDC global interrupt	false	true	true
DMA2D global interrupt	false	true	true
JPEG global interrupt	false	true	true
MDMA global interrupt	false	true	true

*** User modified value**

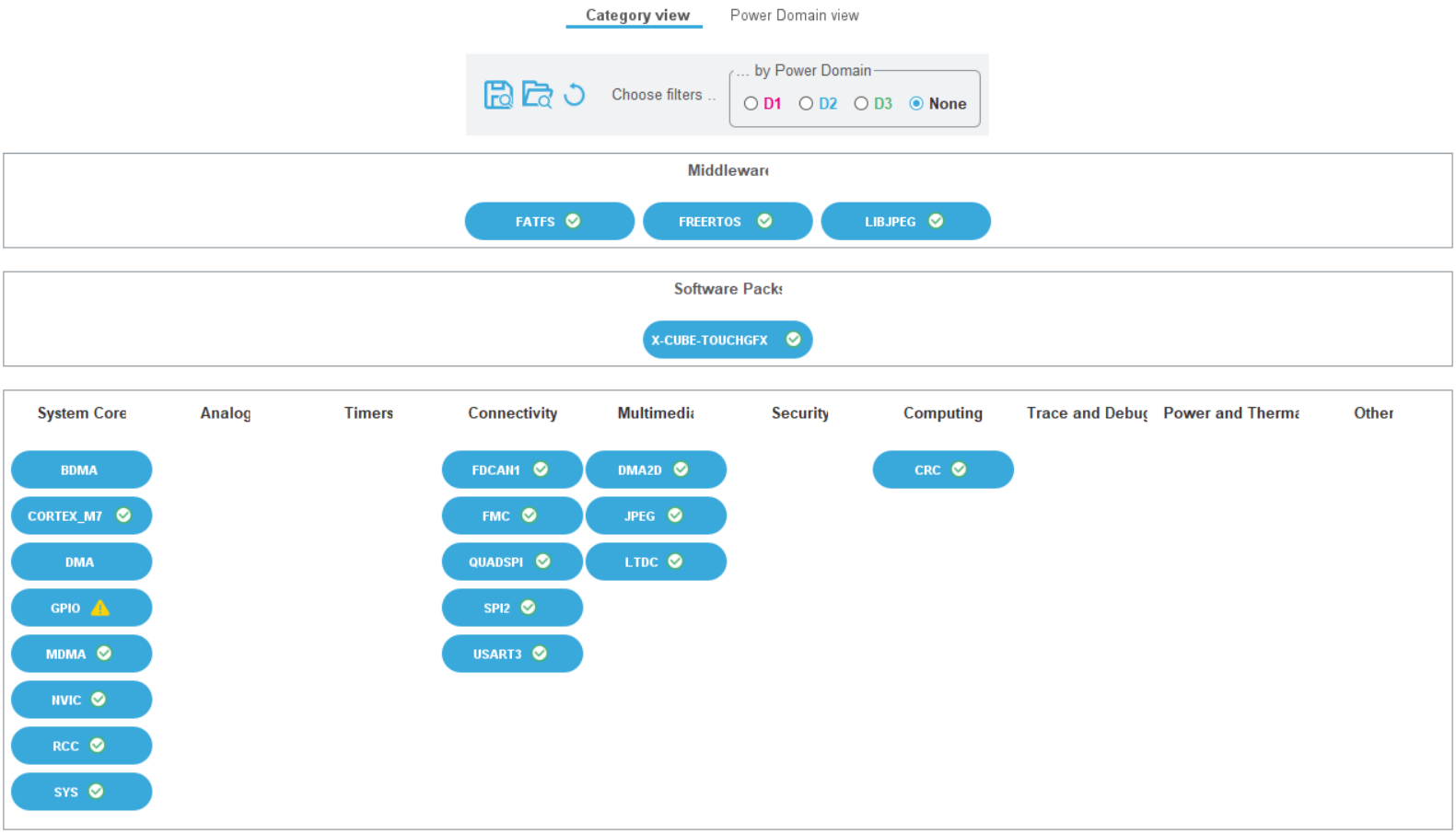
5. System Views

5.1. Category view

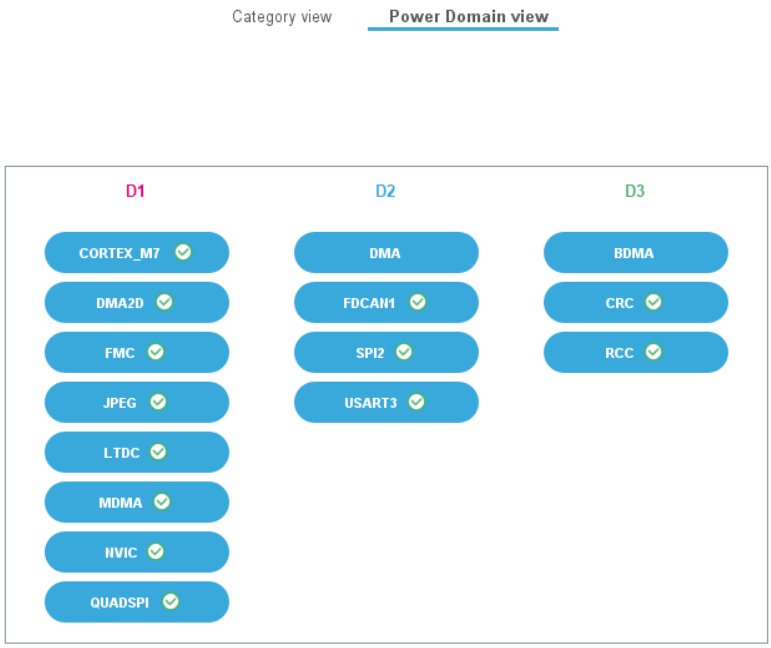
5.1.1. Current



5.1.2. Without filters



5.2. Power Domain view



6. Software Pack Report

6.1. Software Pack selected

Vendor	Name	Version	Component
STMicroelectronics	X-CUBE-TOUCHGFX	4.25.0	Class : Graphics Group : Application Variant : TouchGFX Generator Version : 4.25.0

7. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32f7h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7vl.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-

	stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5188-external-memory-code-execution-on-stm32f7x0-value-line-stm32h750-value-line-stm32h7b0-value-line-and-stm32h730-value-line-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5312-migration-from-revy-to-revv-for-stm32h743753-and-stm32h750-value-line-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5354-getting-started-with-the-stm32h7-series-mcu-16bit-adc-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4760-quadspi-interface-on-stm32-microcontrollers-and-microprocessors--stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4938-getting-started-with-stm32h74xig-and-stm32h75xig-mcu-hardware-development-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4992-introduction-to-secure-firmware-install-sfi-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4861-introduction-to-lcdtft-display-controller-ltdc-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4943-how-to-use-chromart-accelerator-to-refresh-an-lcdtft-display-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5507-how-to-use-crc-to-check-the-integrity-of-the-internal-flash-memory-on-stm32h7-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5996-migrating-from-stm32h730-and-stm32h750-to-stm32h7rx7sx-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5337-guidelines-for-estimating-stm32h7-mcus-lifetime-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4996-how-to-use-the-jpeg-codec-peripheral-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/cd00211314-how-to-

	optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5020-introduction-to-digital-camera-interface-dcmi-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5200-getting-started-with-the-stm32h7-mcu-sdmmc-host-controller-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4323-getting-started-with-stemwin-library-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4435-guidelines-for-obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5001-stm32cube-expansion-package-for-stm32h7-series-mdma-stmicroelectronics.pdf

Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5014-stm32h7x3-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbpd-sink-application-with-stm32cubemx-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4865-lowpower-timer-lptim-applicative-use-cases-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-

for related Tools & Software	and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5450-stm32h7a37b3-lines-and-stm32h7b0-value-line-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6088-how-to-use-mce-for-encryptiondecryption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf
Design Notes & Tips	https://www.st.com/resource/en/design_tip/dt0117-microphone-array-beamforming-in-the-pcm-and-pdm-domain-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0392-stm32h742xig-stm32h743xig-stm32h750xb-stm32h753xi-device-errata-stmicroelectronics.pdf

Datasheet	https://www.st.com/resource/en/datasheet/dm00496677.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0253-stm32f7-series-and-stm32h7-series-cortexm7-processor-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0433-stm32h742-stm32h743753-and-stm32h750-value-line-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um2840-stm32h7-dualcore-series-safety-manual-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um2331-stm32h7-singlecore-series-safety-manual-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3252-stm32h7-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf